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On the Structure and Computation of Extremal Mediated Graphs with Applications to Conic Optimization

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Abstract
In this paper, we introduce a unified definition of *mediated graphs*, a combinatorial structure that underlies several constructions in polynomial and conic optimization. We investigate their geometric and algebraic properties and study extremal mediated graphs under the partial order induced by the cardinality of their vertex sets. We develop mixed-integer linear formulations that enable the exact computation of these graphs, which are typically difficult to obtain by direct enumeration. We show that mediated graphs play a crucial role in SOS and SONC decompositions of polynomials, as well as in second-order cone representations of convex cones, thereby providing new modeling tools with a direct impact on conic optimization. An extensive computational study demonstrates the effectiveness and scalability of the proposed methods.

Keywords Mediated sets · Sum of squares · Sum of nonnegative circuits · Second-order cone · Generalized power cone · Conic optimization · Polynomial optimization · Combinatorial optimization

Mathematics Subject Classification 52B20 · 90C25 · 14P10 · 90C27

1 Introduction

Conic optimization has become a central tool in combinatorial optimization, where conic relaxations frequently yield tighter bounds, improved approximation guarantees, and new structural insights. Semidefinite programming (SDP) plays a particularly prominent role: the Lovász–ϑ function [18] established one of the earliest and most

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influential links between graph invariants and SDP, and subsequentdevelopments consolidated this perspective [11, 28]. SDP relaxations for classical NP-hard problems such as max-cut [12], the quadratic assignment and traveling salesman problems [28], graph partitioning [22], and the stable set problem [10] have demonstrated how SDP-based formulations can substantially strengthen linear models in discrete optimization.

Second-order and p-order cone programming also provide powerful tools for combinatorial modeling, enabling robust or refined formulations in settings such as SOCP-based network design [1, 3] and p-norm relaxations for quadratic optimization [2, 20]. Work on polyhedral approximations of the second-order cone [3] further illustrates the deep geometric-combinatorial interplay inherent in conic relaxations. In several applications, including hub location problems [7] and power-cone representations in conic programming [6], the combinatorial structure of the underlying problem directly interacts with the geometry of the conic model, revealing a rich synergy between discrete and conic optimization frameworks.

A second line of work where conic optimization is essential arises in polynomial optimization. Such problems are nonconvex in general, but convex conic relaxations, most notably the Moment-SOS hierarchy of Lasserre [17], provide systematic ways of approximating polynomial minima over semialgebraic sets. Central to these methods is the ability to certify nonnegativity through sum-of-squares (SOS) representations [23, 24], which convert polynomial constraints into semidefinite constraints. Analogously, the SONC framework and its SOCP-based representations [19] provide alternative nonnegativity certificates for sparse or structured polynomials. These tools have proved influential in optimal control [25], probability, and dynamical systems [14], where verifying stability or safety often reduces to checking polynomial inequalities.

Many of these conic formulations, including SOS, SONC, and power-cone representations, rely on *mediated structures* that relate indices, exponents, or generating points through midpoint relations. This idea originates in the notion of a *mediated set*, introduced by Reznick [27] to characterize when an *agiform* admits an SOS decomposition. Mediated sets later appeared in the construction of SONC cones [13, 15, 16], in SOCP representations of sums of nonnegative circuits [19], in the description of convex semialgebraic sets [9, 29], and in optimal SOCP formulations of weighted geometric mean inequalities [30]. More recently, mediated structures have been used to derive minimal SOCP extended formulations for generalized power cones [5], enabling exact SOCP reformulations of generalized power cone programs.

Across these examples, the same combinatorial object implicitly reappears: a set of points closed under selected midpoint relations, where certain elements depend on the average of two others. However, despite the growing use of such constructions in conic optimization, no systematic *graph-theoretic* or *combinatorial* framework exists to capture, analyze, and compute these structures. Existing approaches treat mediated sets in an ad hoc manner, often with problem-specific constructions that do not generalize or scale.

The contributions of this work can be summarized as follows:

– **A unified combinatorial representation for conic structures.** We introduce *mediated graphs*, a new family of geometric directed graphs embedded in additive abelian groups. These graphs encode midpoint relations between exponent vectors

arising in SOS decompositions of circuit polynomials, and SOCP-based representations of SONC and power cone programs. By identifying the supporting nodes corresponding to monomial degrees and the induced midpoint relations, mediated graphs provide a single combinatorial object that captures the dependency patterns inherent in the interaction between these cones.

- **A clear conceptual framework linking combinatorial and conic optimization.** Many conic representations implicitly rely on discrete midpoint relations, yet these dependencies are typically handled only algebraically. By making them explicit through mediated graphs, we reveal the combinatorial structure that governs the interaction between SOS and SONC cones, or the SOC representation of power cones. This perspective clarifies the role of combinatorial constraints within conic modeling and provides a rigorous foundation for understanding and constructing these representations.
- **A structural theory of mediated graphs.** We develop a systematic analysis of mediated graphs, studying their geometric and algebraic properties and introducing a natural partial order based on vertex sets. Within this order, we characterize *extremal* mediated graphs, minimal and maximal configurations, that correspond to the sparsest and most expressive conic representations, respectively.
- **Mixed-integer optimization models for extremal mediated graphs.** We propose new mixed-integer linear optimization formulations for computing extremal mediated graphs on arbitrary domains. These formulations exploit the structural properties established above and allow the exact computation of minimal and maximal mediated graphs, overcoming the limitations of existing heuristic or enumerative approaches.
- **Applications to conic modeling.** Our framework yields systematic SOCP representations for generalized power cone programs, explicit SOS decompositions of circuit polynomials, and hybrid constructions involving both the SOS and SONC cones. These results provide new tools for combinatorial and polynomial optimization, robust modeling, and conic reformulations of nonlinear constraints.
- **Extensive computational validation.** We conduct a comprehensive computational study evaluating the performance of our mixed-integer formulations, comparing them with state-of-the-art enumerative methods for maximal mediated graphs and investigating continuous and discrete domains for minimal graphs. The experiments demonstrate the scalability, efficiency, and modeling advantages of our approach.

Organization

The remainder of this paper is organized as follows. In Sect. 2, we introduce the notation used throughout the paper, formally define the concept of mediated graphs, and establish some fundamental geometric and algebraic properties derived from their definition. Section 3 is dedicated to studying optimal (extremal) graphs within the class of mediated graphs under a partial order that we define, based on the cardinality of the vertex set. The identification and computation of these optimal graphs form the core of this work. We motivate the need for their efficient computation by highlighting

their connection to the existence of SOS decompositions of circuit polynomials and minimal SOC representations of more complex conic structures. In Sects. 3.1 and 3.2, we introduce and analyze minimal and maximal mediated graphs, respectively. These sections present the main contributions of this work: the development of integer linear optimization (ILO) formulations to compute these graphs. We further illustrate the application of these structures in three different types of problems of interest: representation of generalized power cones, SOS decomposition of sum of nonnegative circuits (SONC), and the analysis of the intersection of the SOS and SONC cones. In Sect. 4, we report the results of an extensive set of experiments demonstrating the validity and practical utility of our approaches. For maximal mediated graphs, we compare our optimization-based method with the enumerative approach proposed in [13]. For minimal mediated graphs, we analyze the performance of different domain formulations for which we derive integer linear optimization models. Finally, we draw some conclusions and indicate some potential future research on the topic.

2 Mediated Graphs

In this section, we introduce the notion of mediated graph and derive some interesting geometric and algebraic properties.

2.1 Notation

Let $G := (V, A)$ be a directed graph, where V is the vertex set and $A \subset V \times V$ is the arc set of G. For each $\boldsymbol{v} \in V$, we denote by $\delta^+(\boldsymbol{v}) := \{\boldsymbol{w} \in V : (\boldsymbol{v}, \boldsymbol{w}) \in A\}$ the set of its outgoing arcs, and $\deg^+(\boldsymbol{v}) := |\delta^+(\boldsymbol{v})|$ is the outdegree (here, $|X|$ denotes the cardinality of the finite set X). Analogously, $\delta^-(\boldsymbol{v}) := \{\boldsymbol{w} \in V : (\boldsymbol{w}, \boldsymbol{v}) \in A\}$ denotes the set of ingoing arcs of $\boldsymbol{v}$, and $\deg^-(\boldsymbol{v}) := |\delta^-(\boldsymbol{v})|$ its indegree. Any directed graph, $G = (V, A)$, has associated three matrices that allow us to characterize the graph, namely the adjacency, the degree, and the Laplacian matrix. The *adjacency matrix* of G is defined as $\mathrm{Adj}(G) := (a_{\boldsymbol{vw}})_{\boldsymbol{v},\boldsymbol{w}\in V}$, where $a_{\boldsymbol{vw}} = 1$ if $(\boldsymbol{v}, \boldsymbol{w}) \in A$, and zero otherwise. The *degree matrix* of G is the diagonal matrix of order $|V| \times |V|$ defined as $D^+(G) := (d_{\boldsymbol{vw}})_{\boldsymbol{v},\boldsymbol{w}\in V}$, where $d_{\boldsymbol{vv}} = \deg^+(\boldsymbol{v})$. Finally, the *Laplacian matrix* of G is defined as $L^+(G) := D^+(G) - \mathrm{Adj}(G)$. Given a subset $U \subset V$, a matrix M_U corresponds with the submatrix of M formed by the rows and columns indexed in U.

Let $\mathbb{R}^d$ be the d-dimensional real vector space, we denote the vectors $\boldsymbol{v} = (v_1, \ldots, v_d) \in \mathbb{R}^d$ by bold characters. Given a subset $\mathbb{M} \subseteq \mathbb{R}^d$, we say that $G = (V, A)$ is said to be an $\mathbb{M}$-geometric digraph if its vertex set, V, is embedded in $\mathbb{M}$, i.e., $V \subset \mathbb{M}$. For a finite set $\mathcal{A} \subset \mathbb{M}$, we denote by $\mathrm{Conv}(\mathcal{A})$ the convex hull of $\mathcal{A}$, and by $\mathrm{Conv}(\mathcal{A})^\circ$ its interior.

We start by introducing the main definition in this work.

Definition 2.1 *(Mediated Graph)* Let $\mathcal{A} \subset \mathbb{M}$ be a finite set. An $\mathbb{M}$-geometric digraph $G = (V, A)$ is said an *$\mathcal{A}$-mediated graph* if $\mathcal{A} \subset V$, $\delta^+(\boldsymbol{v}) \neq \emptyset$, and if $\boldsymbol{w} \in \delta^+(\boldsymbol{v})$, then $2\boldsymbol{v} - \boldsymbol{w} \in \delta^+(\boldsymbol{v})$, for all $\boldsymbol{v} \in V \backslash \mathcal{A}$.

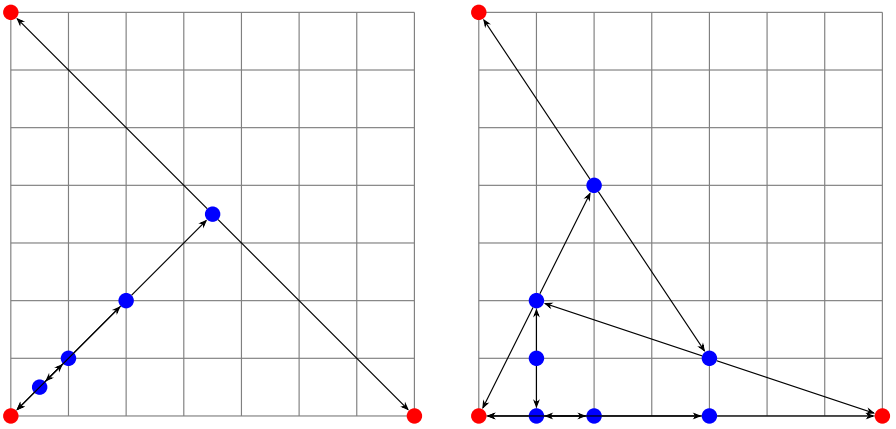


Fig. 1 Two versions of $\mathcal{A}$-mediated graphs but with different vertex domains. $\mathbb{R}^2$ (left) and $\mathbb{Z}^2$ (right). Red dots denote the elements of $\mathcal{A}$, whereas blue dots denote the vertices belonging to the mediated sets but not to $\mathcal{A}$

The family of $\mathcal{A}$-mediated graphs on $\mathbb{M}$ will be denoted by $\mathcal{M}_{\mathcal{A}}^{\mathbb{M}}$. The reference to the domain $\mathbb{M}$ will be omitted in the notation unless it is necessary to avoid confusion.

The definition of a mediated graph implies that every vertex in V, except those in $\mathcal{A}$, is the midpoint of two of its out-adjacent vertices.

Remark 2.1 Note that every nonempty mediated graph $G = (V, A)$ can be reduced to a graph $G' = (V, A')$, whose outdegree $\deg^+(\boldsymbol{v}) = 2$ for all $\boldsymbol{v} \in V \backslash \mathcal{A}$ and $\deg^+(\boldsymbol{a}) = 0$ for all $\boldsymbol{a} \in \mathcal{A}$ which is also a mediated graph. Thus, hereinafter, we assume without loss of generality, that, in every $\mathcal{A}$-mediated graph, each vertex has outdegree 2, except for those in $\mathcal{A}$ which have outdegree 0. Moreover, we assume $\mathcal{A} \neq \emptyset$, since the only $\emptyset$-mediated graph is the empty graph.

In the following example, we illustrate the geometric shapes of some mediated graphs.

Example 2.1 Let $\mathcal{A} = \{(0, 0), (7, 0), (0, 7)\}$ (red dots) in Figure 1 we show two different $\mathcal{A}$-mediated graphs in two different domains. In the left panel, we show an $\mathcal{A}$-mediated graph constructed on $\mathbb{M} = \mathbb{R}^2$. Note that, apart from the points in $\mathcal{A}$, four more vertices appear in the graph (blue dots), namely $(0.5, 0.5)$, $(1, 1)$, $(2, 2)$, and $(3.5, 3.5)$, all of them being midpoints of two other vertices in the graph. The arrows indicate the arcs in the graph, each of them directed from the vertex to the two other vertices for which it is a midpoint. In the right panel, we show an example of an $\mathcal{A}$-mediated graph with domain $\mathbb{M} = \mathbb{Z}^2$. In this case, the graph consists of 10 vertices (3 of them are in $\mathcal{A}$), but the coordinates of the points are now integer numbers. In this case, the vertices not in $\mathcal{A}$ are: $(1, 0)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 4)$, $(4, 0)$, and $(4, 1)$.

Although mediated graphs have been proven to be useful in different fields, in this paper, we provide a general framework for them, as well as explore some of their properties.

2.2 Properties

First, we derive some geometric and algebraic properties of a mediated graph, whose proofs are straightforward, but allow us to understand the closeness properties of the family $\mathcal{M}_{\mathcal{A}}$.

Proposition 2.1 *Let $\mathcal{A} \subset \mathbb{M}$ be a finite set of points in $\mathbb{M}$, and $G = (V, A) \in \mathcal{M}_{\mathcal{A}}$. Then, the following properties are verified:*

1. $G \in \mathcal{M}_{\mathcal{A}}^{\mathbb{M}'}$ *for any* $\mathbb{M} \subseteq \mathbb{M}'$.
2. $G \in \mathcal{M}_{\mathcal{A}'}$ *for all* $\mathcal{A} \subseteq \mathcal{A}' \subset \mathbb{M}$.[1]
3. $\mathcal{M}_{\emptyset} = \{(\emptyset, \emptyset)\}$.
4. *If* $\mathcal{A}' \subset \mathbb{M}$ *and* $(V', A') \in \mathcal{M}_{\mathcal{A}'}$, *then* $(V \cup V', A \cup A') \in \mathcal{M}_{\mathcal{A} \cup \mathcal{A}'}$.
5. $V \subset \mathrm{Conv}(\mathcal{A})$.
6. *If $\mathcal{A}$ is affinely independent and* $V \cap \mathrm{Conv}(\mathcal{A})^{\circ} \neq \emptyset$, *then* $\deg^{-}(\boldsymbol{a}) \geq 1$ *for all* $\boldsymbol{a} \in \mathcal{A}$.
7. *The submatrix $L^{+}(G)_{V \setminus \mathcal{A}}$ of the Laplacian matrix of G is invertible, and the entries of its inverse are nonnegative.*

Proof Since 1-6 are straightforward, we will only prove 7. It is just needed to see if the matrix satisfies the hypothesis of [27, Lemma 4.3]. Since, for all $\boldsymbol{v} \in V \setminus \mathcal{A}$, $\deg^{+}(\boldsymbol{v}) = 2$, thus the entries of the main diagonal of the matrix are equal to two; and it is clear that the entries out of the main diagonal only take values in $\{0, -1\}$. Each row is in the shape $(c_{\boldsymbol{vw}})_{\boldsymbol{w} \in V \setminus \mathcal{A}}$ with $\boldsymbol{v} \in V \setminus \mathcal{A}$, therefore it has as -1's as children of $\boldsymbol{v}$ in $V \setminus \mathcal{A}$ so at most two. Eventually, assume that there exists a principal submatrix in which each row has exactly two -1's. This implies that there exists $U \subseteq V \setminus \mathcal{A}$ such that for all $\boldsymbol{v} \in U$ its two children belong to U. So, the graph $(U, A \cap (U \times U))$ is an $\emptyset$-mediated graph, then $U = \emptyset$ by statement 3. □

In the family of mediated graphs $\mathcal{M}_{\mathcal{A}}$, we will analyze those graphs whose vertex set contains a finite set $\mathcal{B} \subset \mathbb{M}$. These subsets play a main role in this work.

$$\mathcal{M}_{\mathcal{A}}(\mathcal{B}) := \{(V, A) \in \mathcal{M}_{\mathcal{A}} : \mathcal{B} \subset V\}. \tag{1}$$

When $\mathcal{B}$ is a singleton, i.e. $\mathcal{B} = \{\boldsymbol{b}\}$ for some $\boldsymbol{b} \in \mathbb{M}$, $\mathcal{M}_{\mathcal{A}}(\boldsymbol{b}) := \mathcal{M}_{\mathcal{A}}(\{\boldsymbol{b}\})$.

In what follows, we analyze under which conditions the family $\mathcal{M}_{\mathcal{A}}(\mathcal{B})$ is not empty.

Definition 2.2 *(Binary Support)* Let n be a nonnegative integer number. We denote by $\Omega(n)$ the set of strictly positive coefficients in the binary decomposition of n, namely, if $n = \sum_{i=0}^{\lfloor \log_2(n) \rfloor} b_i 2^i$, then

$$\Omega(n) = \left\{ i \in \{0, \ldots, \lfloor \log_2(n) \rfloor\} : b_i = 1 \right\}.$$

We adopt [30, Corollary 28] for mediated graphs to provide a sufficient condition for nonemptiness for $\mathcal{M}_{\mathcal{A}}(\boldsymbol{b})$.

[1] Without considering the isolated vertices in $\mathcal{A}' \setminus \mathcal{A}$.

Lemma 2.1 *Let $\mathcal{A} \subset \mathbb{R}^d$ be an affine independent set and $\boldsymbol{b} \in \mathrm{Conv}(\mathcal{A})^\circ$. Then, there exists a nonempty graph $G = (V, A)$ in $\mathcal{M}_{\mathcal{A}}(\boldsymbol{b})$ with*

$$|V| = \sum_{\boldsymbol{a}\in\mathcal{A}} |\Omega(s_{\boldsymbol{a}})| + |\mathcal{A}| - 1,$$

where $\boldsymbol{s} = (s_{\boldsymbol{a}})_{\boldsymbol{a}\in\mathcal{A}} \in (\mathbb{Z}_+^*)^{|\mathcal{A}|}$, $\gcd(\boldsymbol{s}) = 1$, and $\frac{1}{\|\boldsymbol{s}\|_1}\boldsymbol{s}$ are the barycentric coordinates of $\boldsymbol{b}$ with respect to $\mathcal{A}$.

In the following result, we generalize the previous result for any set $\mathcal{A}$ and any set $\mathcal{B}$ (not necessarily a singleton). Notice that for every point $\boldsymbol{b} \in \mathrm{Conv}(\mathcal{A})$, there exists at least one subset $\mathcal{C} \subseteq \mathcal{A}$ affinely independent (a.i.), strictly containing the point $\boldsymbol{b}$. Hence, we can define

$$\mathcal{A}(\boldsymbol{b}) := \arg\min_{\mathcal{C}\subseteq\mathcal{A}\text{ a.i.}} \left\{ \sum_{\boldsymbol{c}\in\mathcal{C}} |\Omega(s_{\boldsymbol{c}})| + |\mathcal{C}| : \right.$$
$$\left. \boldsymbol{s} = (s_{\boldsymbol{c}})_{\boldsymbol{c}\in\mathcal{C}} \in (\mathbb{Z}_{>0})^{|\mathcal{C}|}, \quad \gcd(\boldsymbol{s}) = 1, \quad \boldsymbol{b} = \frac{1}{\|\boldsymbol{s}\|_1} \sum_{\boldsymbol{c}\in\mathcal{C}} s_{\boldsymbol{c}}\boldsymbol{c} \right\}.$$

We denote by $\boldsymbol{s}^{\boldsymbol{b}} := (s^{\boldsymbol{b}}_{\boldsymbol{a}})_{\boldsymbol{a}\in\mathcal{A}(\boldsymbol{b})} \in (\mathbb{Z}_+^*)^{|\mathcal{A}(\boldsymbol{b})|}$, where $\gcd(\boldsymbol{s}^{\boldsymbol{b}}) = 1$ and $\frac{1}{\|\boldsymbol{s}^{\boldsymbol{b}}\|_1}\boldsymbol{s}^{\boldsymbol{b}}$ are the barycentric coordinates of $\boldsymbol{b}$ with respect to $\mathcal{A}(\boldsymbol{b})$.

Theorem 2.1 *Let $\mathcal{A}, \mathcal{B} \subset \mathbb{R}^d$ be finite subsets and $\mathcal{B} \subset \mathrm{Conv}(\mathcal{A})$. Then, there exists a nonempty graph $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$ such that:*

$$|V| \le \nu_{\mathcal{A}}(\mathcal{B}) := \sum_{\boldsymbol{b}\in\mathcal{B}} \left[\sum_{\boldsymbol{a}\in\mathcal{A}(\boldsymbol{b})} |\Omega(s^{\boldsymbol{b}}_{\boldsymbol{a}})| + |\mathcal{A}(\boldsymbol{b})| \right] - |\mathcal{B}|.$$

Proof By Lemma 2.1, there exists $G_{\mathcal{A}(\boldsymbol{b})} = (V_{\mathcal{A}(\boldsymbol{b})}, A_{\mathcal{A}(\boldsymbol{b})})$ in $\mathcal{M}_{\mathcal{A}(\boldsymbol{b})}(\boldsymbol{b}) \subseteq \mathcal{M}_{\mathcal{A}}(\boldsymbol{b})$ (the inclusion is obtained from Proposition 2.1.2) with

$$|V_{\mathcal{A}(\boldsymbol{b})}| = \sum_{\boldsymbol{a}\in\mathcal{A}(\boldsymbol{b})} |\Omega(s^{\boldsymbol{b}}_{\boldsymbol{a}})| + |\mathcal{A}(\boldsymbol{b})| - 1.$$

By Proposition 2.1.4, there must be a graph $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$ where $V = \bigcup_{\boldsymbol{b}\in\mathcal{B}} V_{\mathcal{A}(\boldsymbol{b})}$, thus $|V| \le \sum_{\boldsymbol{b}\in\mathcal{B}} |V_{\mathcal{A}(\boldsymbol{b})}|$. □

Note that the nonemptiness of $\mathcal{M}_{\mathcal{A}}(\mathcal{B})$ is not assured in general, even if $\mathcal{B} \subset \mathrm{Conv}(\mathcal{A}) \cap \mathbb{M}$. In the following result, we show a counterexample of this situation.

Example 2.2 Let us consider $\mathbb{M} = (2\mathbb{Z})^2$, $\mathcal{A} = \{(0, 0), (4, 2), (2, 4)\}$ and $\boldsymbol{b} = (2, 2) \in \mathrm{Conv}(\mathcal{A}) \cap (2\mathbb{Z})^2$. Note that there is no way to construct an $\mathcal{A}$-mediated graph that contains $\boldsymbol{b}$, and then $\mathcal{M}_{\mathcal{A}}(\boldsymbol{b}) = \emptyset$.

In view of the possible nonexistence of mediated graphs, the authors in [26] provide a lower bound for the *dilation* constant $k \in \mathbb{Z}_+$ such that $\mathcal{M}_{k\mathcal{A}}(\mathcal{B})$ is nonempty for all $\mathcal{A} \subset \mathbb{M}$, where $k\mathcal{A} := \{k\boldsymbol{a} \in (2\mathbb{Z})^d : \boldsymbol{a} \in \mathcal{A}\}$ and $\mathcal{B} \subset \text{Conv}(k\mathcal{A}) \cap \mathbb{Z}^d$ in the case $\mathbb{M} = (2\mathbb{Z})^d$.

Theorem 2.2 *Let $\mathcal{A} \subset (2\mathbb{Z})^d$ be an affinely independent set. Then, for every integer $k \geq \max\{2, |\mathcal{A}|-2\}$, there exists an $(k\mathcal{A})$-mediated graph, $G = (\text{Conv}(k\mathcal{A}) \cap \mathbb{Z}^d, A)$ such that $\delta^+(\boldsymbol{v}) \subset (2\mathbb{Z})^d$ for every vertex $\boldsymbol{v} \in \text{Conv}(k\mathcal{A}) \cap \mathbb{Z}^d$.*

Proof The claim follows directly from Definition 2.1 and [26, Theorem 3.1]. □

3 Optimal Mediated Graphs

The family of $\mathcal{A}$-mediated graphs can be endowed with a partial order induced by the cardinality of their vertex sets. Specifically, given $G = (V, A)$ and $G' = (V', A') \in \mathcal{M}_{\mathcal{A}}$:

$$G \preceq G' \text{ if either } |V| < |V'| \text{ or } G = G'.$$

Based on this partial order, the family of $\mathcal{A}$-mediated graphs for a given finite set $\mathcal{A} \subset \mathbb{M}$ has two distinguished elements that arise when minimizing and maximizing the number of vertices in the graph. Since the enumeration of this family of graphs can be, in general, cumbersome, we develop approaches to construct these *optimal* graphs using mathematical optimization tools, which, as already announced, constitute the main contribution of this paper. The study and construction of these so-called minimal and maximal mediated graphs are motivated by their implications in different problems in convex algebraic geometry, as well as in their application to efficiently represent useful convex optimization problems.

We separately analyze minimal and maximal mediated graphs for different reasons. On the one hand, Minimal Mediated Graphs (MinMG, for short) have already been proven to be useful structures to derive the most efficient and simplest SOC or SOS representations of certain more complex convex sets, with a high impact in practical convex optimization problems (see e.g., [5, 8, 30]). Furthermore, MinMG can be analyzed in *discrete* and *continuous* domains, referring to the cardinality of $\text{Conv}(\mathcal{A}) \cap \mathbb{M}$, which may be finite or infinite, respectively. On the other hand, the known applications of Maximal Mediated Graphs (MaxMG, for short) are related to the existence of SOS decompositions of circuit polynomials [13, 27]. Clearly, the construction of MaxMG is only possible in discrete domains.

In this section, we analyze some properties of MinMG and MaxMG and propose different approaches for their construction.

3.1 Minimal Mediated Graphs

Note that the unique minimal mediated graph of $\mathcal{M}_{\mathcal{A}}$ is the edgeless graph with vertex set $\mathcal{A}$. Thus, we analyze here MinMG restricted to the family $\mathcal{M}_{\mathcal{A}}(\mathcal{B})$ for a nonempty set $\mathcal{B} \subset \text{Conv}(\mathcal{A}) \cap \mathbb{M}$ (see (1)). In what follows, we formally introduce this family of subgraphs.

Definition 3.1 *(Minimal Mediated Graph)* Let $\mathcal{A}, \mathcal{B} \subset \mathbb{M}$ be finite sets. We say that $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$ is a *minimal $\mathcal{A}$-mediated graph* for $\mathcal{B}$ if there does not exist $G' = (V', A') \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$ such that $|V'| < |V|$.

We denote by $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$ the family of minimal $\mathcal{A}$-mediated graphs for $\mathcal{B}$. Notice that a minimal mediated graph is a minimal element in the poset $(\mathcal{M}_{\mathcal{A}}(\mathcal{B}), \preceq)$.

Some properties can be derived for this family of graphs.

Theorem 3.1 *Let $\mathcal{A} \subset \mathbb{M}$ and $\mathcal{B} \subset \mathbb{M}$. Then any minimal mediated graph $G \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$ has at most $|\mathcal{B}|$ weakly connected components.*

Proof Let us first analyze the case when $\mathcal{B} = \{\boldsymbol{b}\}$. In case $\mathcal{M}_{\mathcal{A}}(\boldsymbol{b}) = \emptyset$, the result is clear. We prove the result by contraposition for nonempty $\mathcal{M}_{\mathcal{A}}(\boldsymbol{b})$. Suppose the graph $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\boldsymbol{b})$ is not weakly connected. Thus, one can construct the graph $G' = (V', A')$ where

$$V' := \{\boldsymbol{v} \in V : \text{there exists a path from } \boldsymbol{b} \text{ to } \boldsymbol{v}\}, \; A' := A \cap (V' \times V').$$

Note that G' is an $\mathcal{A}$-mediated graph with $\boldsymbol{b} \in V'$. Given $\boldsymbol{v} \in V' \backslash \mathcal{A}$, by definition, there exists a path in G from $\boldsymbol{b}$ to $\boldsymbol{v}$. If $\boldsymbol{w}$ and $2\boldsymbol{v} - \boldsymbol{w}$ are the children of $\boldsymbol{v}$ in G, then $\boldsymbol{w}, 2\boldsymbol{v} - \boldsymbol{w} \in V'$, and therefore $(\boldsymbol{v}, \boldsymbol{w})$ and $(\boldsymbol{v}, 2\boldsymbol{v} - \boldsymbol{w})$ belong to A'. Since $\boldsymbol{b} \in V'$, we have $G' \in \mathcal{M}_{\mathcal{A}}(\boldsymbol{b})$. As G is not weakly connected, $V' \subsetneq V$, hence $|V'| < |V|$, contradicting the minimality of G.

Let us now analyze the general case. If $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$ is not weakly connected, it is enough to consider the graph $G'' = (V'', A'')$ defined by

$$V'' := \{\boldsymbol{v} \in V : \text{there exists a path from some vertex of } \mathcal{B} \text{ to } \boldsymbol{v}\},$$
$$A'' := A \cap (V'' \times V''),$$

which, analogously to the singleton case, belongs to $\mathcal{M}_{\mathcal{A}}(\mathcal{B})$, with $|V''| < |V|$, contradicting the minimality of G. □

The following result is straightforward from the construction in the previous proof and provides information about the indegrees of the vertices in a minimal mediated graph.

Corollary 3.1 *If $G \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$ is a minimal mediated graph, then $\deg^-(\boldsymbol{v}) \geq 1$ for all $\boldsymbol{v} \in V \backslash \mathcal{B}$.*

The next result is a straightforward consequence of Theorem 2.1, giving an upper bound for the cardinality of a minimal mediated graph.

Corollary 3.2 *Let $\mathcal{A}, \mathcal{B} \subset \mathbb{R}^d$ and $G = (V, A) \in \text{MinMG}_{\mathcal{A}}(\mathcal{B})$. Then*

$$|V| \leq \nu_{\mathcal{A}}(\mathcal{B}).$$

Computing $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$ requires determining both the vertices in $\text{Conv}(\mathcal{A})$ and the arcs that satisfy the *mediated* condition with minimum cardinality. For particularly

structured sets $\mathcal{A}$ and singleton sets $\mathcal{B}$, [30] proposes a brute-force algorithm and different heuristic approaches to compute $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$, focused on deriving minimal SOCP representations of weighted geometric mean inequalities. [5] provides a mixed-integer linear programming model (MILP) to compute this set. In both cases, $\mathbb{M} = \mathbb{R}^d$ for a given dimension $d \geq 1$. This framework will be referred to in this work as the *continuous-domain* minimal mediated graph, since the vertices are to be found in the continuous region $\text{Conv}(\mathcal{A})$. On the other hand, in the case where $\mathbb{M}$ is a *lattice* (such as $\mathbb{Z}^d$ or $(2\mathbb{Z})^d$), we call this framework the *discrete-domain* minimal mediated set. The main difference lies in the fact that the search for the vertices and arcs in the continuous-domain case must be performed among infinitely many vectors, whereas in the discrete-domain case, the search space is finite. By analogy with facility location theory, this distinction parallels the difference between continuous and discrete location problems.

In what follows, we analyze minimal mediated sets for these two families of domains and derive mathematical optimization models that allow us to compute the minimal mediated graphs, thus avoiding, in general, the enumeration of all potential minimal subgraphs.

3.1.1 Minimal Mediated Graphs in $\mathbb{R}^d$

The problem of deriving mathematical optimization models for the continuous case was already addressed in [5] for the vertex sets of dilations of the standard simplex and a single interior lattice point. Below, we state the result for the general continuous-domain case, which is a straightforward extension of the formulation provided in the aforementioned paper.

Theorem 3.2 *Let $\mathcal{A} \subset \mathbb{R}^d$ be a finite set of points and $\mathcal{B} \subset \text{Conv}(\mathcal{A})$ a nonempty set. The following mixed-integer linear programming model allows one to compute $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$ for $\mathbb{M} = \mathbb{R}^d$.*

(C-MinMGP)

$$\text{Minimize} \quad |\mathcal{A}| + |\mathcal{B}| + \sum_{\boldsymbol{v} \in \mathcal{V}\setminus(\mathcal{A}\cup\mathcal{B})} z_{\boldsymbol{v}} \tag{2}$$

$$\text{subject to} \quad y_{\boldsymbol{vw}} \leq z_{\boldsymbol{v}}, \qquad \forall \boldsymbol{v} \in \mathcal{V}\setminus\mathcal{A}, \boldsymbol{w} \in \mathcal{V}, \tag{3}$$

$$y_{\boldsymbol{vw}} \leq z_{\boldsymbol{w}}, \qquad \forall \boldsymbol{v} \in \mathcal{V}\setminus\mathcal{A}, \boldsymbol{w} \in \mathcal{V}, \tag{4}$$

$$2\boldsymbol{x}_{\boldsymbol{v}} \geq \boldsymbol{x}_{\boldsymbol{w}} + \boldsymbol{x}_{\boldsymbol{u}} - \Delta(2 - y_{\boldsymbol{vw}} - y_{\boldsymbol{vu}}), \qquad \forall \boldsymbol{v} \in \mathcal{V}\setminus\mathcal{A}, \boldsymbol{w} \neq \boldsymbol{u} \in \mathcal{V}, \tag{5}$$

$$2\boldsymbol{x}_{\boldsymbol{v}} \leq \boldsymbol{x}_{\boldsymbol{w}} + \boldsymbol{x}_{\boldsymbol{u}} + \Delta(2 - y_{\boldsymbol{vw}} - y_{\boldsymbol{vu}}), \qquad \forall \boldsymbol{v} \in \mathcal{V}\setminus\mathcal{A}, \boldsymbol{w} \neq \boldsymbol{u} \in \mathcal{V}, \tag{6}$$

$$\|\boldsymbol{x}_{\boldsymbol{v}} - \boldsymbol{x}_{\boldsymbol{w}}\|_1 \geq \varepsilon(z_{\boldsymbol{v}} + z_{\boldsymbol{w}} - 1), \qquad \forall \boldsymbol{v} \neq \boldsymbol{w} \in \mathcal{V}, \tag{7}$$

$$\sum_{\boldsymbol{w} \in \mathcal{V}} y_{\boldsymbol{vw}} = 2z_{\boldsymbol{v}}, \qquad \forall \boldsymbol{v} \in \mathcal{V}\setminus\mathcal{A}, \tag{8}$$

$$\boldsymbol{x}_{\boldsymbol{v}} = \boldsymbol{v}, \quad z_{\boldsymbol{v}} = 1, \qquad \forall \boldsymbol{v} \in \mathcal{A} \cup \mathcal{B}, \tag{9}$$

$$y_{\boldsymbol{vw}} \in \{0, 1\}, \qquad \forall \boldsymbol{v} \in \mathcal{V}\setminus\mathcal{A}, \boldsymbol{w} \in \mathcal{V},$$

$$z_{\boldsymbol{v}} \in \{0, 1\}, \qquad \forall \boldsymbol{v} \in \mathcal{V},$$

$$\boldsymbol{x}_{\boldsymbol{v}} \in \mathbb{R}^d, \qquad \forall \boldsymbol{v} \in \mathcal{V},$$

where $\mathcal{V}$ is any index set satisfying $\mathcal{A}, \mathcal{B} \subset \mathcal{V}$, and $|\mathcal{V}| = v_{\mathcal{A}}(\mathcal{B})$. The parameters $\Delta := \max\{\|\boldsymbol{a}\|_\infty : \boldsymbol{a} \in \mathcal{A}\}$ and sufficiently small tolerance ε is a

Proof In the above formulation, the following variables are used:

$$z_{\boldsymbol{v}} = \begin{cases} 1 & \text{if } \boldsymbol{v} \text{ is a vertex of the minimal mediated graph,} \\ 0 & \text{otherwise;} \end{cases}$$

$$y_{\boldsymbol{v}\boldsymbol{w}} = \begin{cases} 1 & \text{if } (\boldsymbol{v}, \boldsymbol{w}) \text{ is an arc of the minimal mediated graph,} \\ 0 & \text{otherwise;} \end{cases}$$

$\boldsymbol{x}_{\boldsymbol{v}} \in \mathbb{R}^d$: coordinates of the vertices in the MinMG in case $z_{\boldsymbol{v}} = 1, \forall \boldsymbol{v} \in \mathcal{V}$.

Constraints (3) and (4) determine that an arc in the mediated graph is possible only when both endpoints are vertices of the graph. Constraints (5) and (6) ensure the correct construction of a mediated graph, that is, if $(\boldsymbol{v}, \boldsymbol{w})$ and $(\boldsymbol{v}, \boldsymbol{u})$ are arcs in the mediated graph, then $\boldsymbol{v}$ is the midpoint of $\boldsymbol{w}$ and $\boldsymbol{u}$. Constraint (7) ensures that each vertex, seen as an embedded coordinate in $\mathbb{R}^d$, is activated only once. Constraint (8) is derived from Remark 2.1, where the mediated graph is simplified to one with exactly two outgoing arcs from each *proper* mediated vertex. Finally, Constraint (9) sets as vertices of the mediated graph those in $\mathcal{A} \cup \mathcal{B}$. The minimality is ensured by the minimization criterion on the number of mediated vertices (2). □

Note that the Manhattan norm constraints (7) are nonlinear, but they can be rewritten in a standard way as linear constraints. Thus, the model above results in a MILP problem.

Optimization over Generalized Power Cones

Minimal mediated graphs in $\mathbb{R}^d$ have a direct impact on the solution of conic optimization problems. More specifically, a generalized power cone program has the following shape:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^T \boldsymbol{x} + \boldsymbol{d}^T \boldsymbol{z} \\ \text{subject to} \quad & A\boldsymbol{x} + B\boldsymbol{z} = \boldsymbol{b}, \\ & \|\boldsymbol{x}\|_p \le \boldsymbol{z}^{\boldsymbol{\alpha}} \\ & \boldsymbol{x} \in \mathbb{R}^n, \\ & \boldsymbol{z} \in \mathbb{R}^d_+, \end{aligned} \tag{GPCP}$$

where $p \in \mathbb{R}_{\ge 1}$, $\boldsymbol{\alpha} \in \Lambda_d := \{(\alpha_1, \ldots, \alpha_d) \in \mathbb{R}^+ : \sum_{j=1}^d \alpha_j = 1\}$ (d-dimensional standard simplex), $\|\cdot\|_p$ denotes the ℓ_p-norm in $\mathbb{R}^n$, and for any $\boldsymbol{z} \in \mathbb{R}^d_+$ and $\boldsymbol{\alpha} \in \Lambda_d$, $\boldsymbol{z}^{\boldsymbol{\alpha}} = \prod_{j=1}^d z_j^{\alpha_j} = z_1^{\alpha_1} \cdots z_d^{\alpha_d}$ is the weighted geometric mean of $\boldsymbol{z}$ with weights $\boldsymbol{\alpha}$.

The above problem, although it belongs to the family of convex conic problems, needs to be equivalently reformulated as a second-order cone problem in order to

be numerically solved in practice (e.g., using available off-the-shelf solvers). Using solutions of C-MinMGP, one can build a minimal extended reformulation of a GPCP as a SOCP. Let $\mathcal{A}_m := \{\boldsymbol{e}_i : i = 1, \ldots, m\} \cup \{\boldsymbol{0}\} \subset \mathbb{R}^m$ (where $\boldsymbol{e}_i$ stands for the ith unit vector), $\beta_p := \frac{1}{p}$, and $\boldsymbol{\beta}_{\boldsymbol{\alpha}} := (\alpha_1, \ldots, \alpha_{d-1})$. Then each pair of mediated graphs $G_p = (V_p, A_p) \in \mathcal{M}_{\mathcal{A}_1}(\beta_p)$ and $G_{\boldsymbol{\alpha}} = (V_{\boldsymbol{\alpha}}, A_{\boldsymbol{\alpha}}) \in \mathcal{M}_{\mathcal{A}_{d-1}}(\boldsymbol{\beta}_{\boldsymbol{\alpha}})$ is in one-to-one correspondence with the following SOCP extended reformulation of GPCP (see [5] for further details):

$$
\begin{aligned}
\text{minimize} \quad & \boldsymbol{c}^T \boldsymbol{x} + \boldsymbol{d}^T \boldsymbol{z} \\
\text{subject to} \quad & A\boldsymbol{x} + B\boldsymbol{z} = \boldsymbol{b}, \\
& \sum_{j=1}^{d} t_j \le \eta_{\boldsymbol{\beta}_{\boldsymbol{\alpha}}}, \\
& \xi_{j\beta_p} = x_j, \quad \xi_{j1} = t_j, \quad \xi_{j0} = \eta_{\boldsymbol{\beta}_{\boldsymbol{\alpha}}}, && j = 1, \ldots, n, \\
& \eta_{\boldsymbol{e}_i} = z_i, \quad \eta_{\boldsymbol{0}} = z_d, && i = 1, \ldots, d-1, \\
& \xi_{jv}^2 \le \prod_{u \in \delta_p^+(v)} \xi_{ju}, && j = 1, \ldots, n;\ v \in V_p \backslash \mathcal{A}_p, \\
& \eta_{\boldsymbol{v}}^2 \le \prod_{\boldsymbol{u} \in \delta_{\boldsymbol{\alpha}}^+(\boldsymbol{v})} \eta_{\boldsymbol{u}}, && \boldsymbol{v} \in V_{\boldsymbol{\alpha}} \backslash \mathcal{A}_{\boldsymbol{\alpha}}, \\
& \boldsymbol{t}, \boldsymbol{z}, \boldsymbol{\xi}, \boldsymbol{\eta} \ge 0.
\end{aligned}
$$

The number of SOC constraints in the reformulation is $n(|V_p| - 2) + |V_{\boldsymbol{\alpha}}| - d$, the number of variables is $n|V_p| + |V_{\boldsymbol{\alpha}}|$, and the number of linear constraints is $O(m + n|V_p| + |V_{\boldsymbol{\alpha}}|)$ taking into account the ones hiding in the SOC constraints. According to [4], assuming that the coefficients of the input data $(A, B, \boldsymbol{b}, \boldsymbol{c}, \boldsymbol{d})$ have bit size at most τ, then the feasibility of GPCP can be tested in $m(n|V_p| + |V_{\boldsymbol{\alpha}}|)^{O(n|V_p| + |V_{\boldsymbol{\alpha}}|)}$ arithmetic operations over $\tau(n|V_p| + |V_{\boldsymbol{\alpha}}|)^{O(n|V_p| + |V_{\boldsymbol{\alpha}}|)}$-bit numbers. An ε-optimal solution of GPCP can be obtained through binary search in $(\tau + N)m(n|V_p| + |V_{\boldsymbol{\alpha}}|)^{O(n|V_p| + |V_{\boldsymbol{\alpha}}|)}$ arithmetic operations, where $N = 2^{-\varepsilon} \in \mathbb{Z}_+$. Therefore, finding minimal mediated graphs is highly recommended to reduce the computational complexity of GPCP.

On the other hand, note that the solution of the problem allows us to construct the minimal mediated graph, and then, the explicit SOCP representation of the problem. Specifically, let $(\boldsymbol{x}_p^*, \boldsymbol{z}_p^*, \boldsymbol{y}_p^*)$ be the solution of C-MinMGP for $\mathcal{A}_1$ and $\{\beta_p\}$, defining $V_p^* := \{x_{pv}^* \in \mathbb{R} : z_{pv}^* = 1\}$ and $A_p^* := \{(x_{pv}^*, x_{pu}^*) \in \mathbb{R}^2 : y_{pvu}^* = 1\}$, then by Theorem 3.2, the mediated graph $G_p^* := (V_p^*, A_p^*) \in \mathrm{MinMG}_{\mathcal{A}_1}(\beta_p)$. Likewise, let $(\boldsymbol{x}_{\boldsymbol{\alpha}}^*, \boldsymbol{z}_{\boldsymbol{\alpha}}^*, \boldsymbol{y}_{\boldsymbol{\alpha}}^*)$ be the solution of C-MinMGP for $\mathcal{A}_{d-1}$ and $\{\boldsymbol{\beta}_{\boldsymbol{\alpha}}\}$, defining $V_{\boldsymbol{\alpha}}^* := \{\boldsymbol{x}_{\boldsymbol{\alpha}\boldsymbol{v}}^* \in \mathbb{R}^{d-1} : z_{\boldsymbol{\alpha}\boldsymbol{v}}^* = 1\}$ and $A_{\boldsymbol{\alpha}}^* := \{(\boldsymbol{x}_{\boldsymbol{\alpha}\boldsymbol{v}}^*, \boldsymbol{x}_{\boldsymbol{\alpha}\boldsymbol{u}}^*) \in \mathbb{R}^{d-1} \times \mathbb{R}^{d-1} : y_{\boldsymbol{\alpha}\boldsymbol{v}\boldsymbol{u}}^* = 1\}$, then by Theorem 3.2, the mediated graph $G_{\boldsymbol{\alpha}}^* := (V_{\boldsymbol{\alpha}}^*, A_{\boldsymbol{\alpha}}^*) \in \mathrm{MinMG}_{\mathcal{A}_{d-1}}(\boldsymbol{\beta}_{\boldsymbol{\alpha}})$. Hence, the SOCP defined by G_p^* and $G_{\boldsymbol{\alpha}}^*$ is a minimal SOCP extended reformulation of GPCP, which is also SDP-representable.

Similar applications of Theorem 3.2 can be used to derive optimal SOCP reformulations in SONC optimization, matrix optimization, or quantum information. The reader is referred to [19, 30] for further details.

3.1.2 Minimal Mediated Graphs in Discrete Domains

The computation of discrete-domain minimal mediated sets, as far as we know, has not been previously addressed, and although the above model could be adapted (by enforcing that the $\boldsymbol{x}$-variables can only take feasible values in the finite set of points inside $\text{Conv}(\mathcal{A}) \cap \mathbb{Z}^d$ or $\text{Conv}(\mathcal{A}) \cap (2\mathbb{Z})^d$), it would not exploit the discrete nature of the problem and would result in an inefficient approach to compute MinMG. In what follows, we describe an alternative integer linear programming model (ILP) that we propose for this problem.

Theorem 3.3 *Let $\mathcal{A}, \mathcal{B} \subset \mathbb{Z}^d$ be two finite sets of points with $\mathcal{B} \subset \text{Conv}(\mathcal{A})$. The following integer linear programming model allows one to compute a mediated graph in $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$.*

(D-MinMGP)

$$\textit{Minimize} \quad |\mathcal{A}| + |\mathcal{B}| + \sum_{\boldsymbol{v} \in \mathcal{V} \setminus (\mathcal{A} \cup \mathcal{B})} x_{\boldsymbol{v}} \tag{10}$$

$$\textit{subject to} \quad y_{\boldsymbol{vw}} \leq x_{\boldsymbol{v}}, \qquad \forall \boldsymbol{v}, \boldsymbol{w} \in \mathcal{V}, \tag{11}$$

$$y_{\boldsymbol{vw}} \leq x_{\boldsymbol{w}}, \qquad \forall \boldsymbol{v}, \boldsymbol{w} \in \mathcal{V}, \tag{12}$$

$$y_{\boldsymbol{v}(2\boldsymbol{v}-\boldsymbol{w})} \geq y_{\boldsymbol{vw}}, \qquad \forall \boldsymbol{v} \neq \boldsymbol{w} \in \mathcal{V} \textit{ if } 2\boldsymbol{v} - \boldsymbol{w} \in \mathcal{V}, \tag{13}$$

$$y_{\boldsymbol{vw}} = 0, \qquad \forall \boldsymbol{v} \neq \boldsymbol{w} \in \mathcal{V} \textit{ if } 2\boldsymbol{v} - \boldsymbol{w} \notin \mathcal{V}, \tag{14}$$

$$\sum_{\boldsymbol{w} \in \mathcal{V}} y_{\boldsymbol{vw}} = 2x_{\boldsymbol{v}}, \qquad \forall \boldsymbol{v} \in \mathcal{V} \setminus \mathcal{A}, \tag{15}$$

$$x_{\boldsymbol{v}} = 1, \qquad \forall \boldsymbol{v} \in \mathcal{A} \cup \mathcal{B}, \tag{16}$$

$$x_{\boldsymbol{v}} \in \{0, 1\}, \qquad \forall \boldsymbol{v} \in \mathcal{V}, \tag{17}$$

$$y_{\boldsymbol{vw}} \in \{0, 1\}, \qquad \forall \boldsymbol{v}, \boldsymbol{w} \in \mathcal{V}. \tag{18}$$

where $\mathcal{V}$ is the (finite) set of potential positions for the mediated vertices.

Proof Defining the binary variables that completely identify a graph in $\mathcal{M}_{\mathcal{A}}(\mathcal{B})$:

$$x_{\boldsymbol{v}} = \begin{cases} 1 & \text{if } \boldsymbol{v} \text{ is a mediated vertex in } \text{MinMG}_{\mathcal{A}}(\mathcal{B}), \\ 0 & \text{otherwise;} \end{cases}$$

$$y_{\boldsymbol{vw}} = \begin{cases} 1 & \text{if } (\boldsymbol{v}, \boldsymbol{w}) \text{ is an arc in } \text{MinMG}_{\mathcal{A}}(\mathcal{B}), \\ 0 & \text{otherwise.} \end{cases}$$

These variables are properly defined by Constraints (11)–(18). Constraints (11) and (12) ensure that no arcs are possible unless the endpoint vertices are part of the graph. Constraints (13) and (15) imply the verification of the *mediated* condition: if $(\boldsymbol{v}, \boldsymbol{w})$ is an arc in the mediated graph, then $(\boldsymbol{v}, 2\boldsymbol{v} - \boldsymbol{w})$ is also an arc, provided that both endpoints are feasible vertices. Constraint (14) further states that all mediated vertices (except those in $\mathcal{A}$) must have exactly two outgoing arcs. Constraint (11) and (12)

enforce the consistency of vertex selection and associated arcs. The domains of the variables are given in Constraints (17) and (18). The minimality is ensured by the minimization criterion on the number of mediated vertices (10). Note that the elements of $\mathcal{A}$ are excluded from the sum in the objective function, since they are always present in the mediated graph; therefore, their contribution is incorporated as a constant term (which may be omitted when solving the model). □

Remark 3.1 The ILP model in the previous result has $O(|\mathcal{V}|^2)$ variables and $O(|\mathcal{V}|^2)$ linear constraints. For large $\mathcal{V}$, the problem can be computationally costly. Some strategies can be applied to the model in order to facilitate the solution procedure:

- Constraints (11) are not necessary. They are already induced by Constraints (15). Indeed, when $x_{\boldsymbol{v}} = 0$, all outgoing arcs from $\boldsymbol{v}$ are not allowed in the solution.
- Constraints (12) can be strengthened by aggregating all the constraints for a given $\boldsymbol{w}$, i.e., one can replace Constraints (12) by

$$\sum_{\boldsymbol{v}\in\mathcal{V}} y_{\boldsymbol{v}\boldsymbol{w}} \leq |\mathcal{V}| x_{\boldsymbol{w}}, \qquad \forall \boldsymbol{w} \in \mathcal{V}.$$

- The y-variables do not need to be defined for the case $y_{\boldsymbol{v}\boldsymbol{v}}$ with $\boldsymbol{v} \in \mathcal{V}$, since loops are not permitted in a mediated graph. The same applies to the $x_{\boldsymbol{v}}$-variables for $\boldsymbol{v} \in \mathcal{A} \cup \mathcal{B}$, which are already fixed to one in our model.

Remark 3.2 Note that the ILP presented above allows us to compute a single minimal mediated graph in the discrete-domain case. It may be possible, however, that more than one mediated graph attains this minimum cardinality. One can obtain the entire set of mediated graphs in $\mathrm{MinMG}_{\mathcal{A}}(\mathcal{B})$ by iteratively solving the problem and adding constraints to *filter out* the previously obtained mediated graphs until the optimal solution achieves an objective value strictly larger than the minimum cardinality.

Specifically, suppose that solving the problem for the first time yields a minimum cardinality of ν, with the corresponding set of arcs $A_{(1)}$ (given by the y-variables). One may then add the constraints

$$|\mathcal{A}| + |\mathcal{B}| + \sum_{\boldsymbol{v}\in\mathcal{V}\setminus(\mathcal{A}\cup\mathcal{B})} x_{\boldsymbol{v}} \leq \nu,$$
$$\sum_{\boldsymbol{v},\boldsymbol{w}\in\mathcal{V}} y_{\boldsymbol{v}\boldsymbol{w}} \leq |A_{(1)}| - 1,$$

and solve the problem again. Thus, in the kth iteration, when the set of arcs $A_{(k)}$ is obtained, we add the constraint

$$\sum_{\boldsymbol{v},\boldsymbol{w}\in\mathcal{V}} y_{\boldsymbol{v}\boldsymbol{w}} \leq |A_{(k)}| - 1.$$

If the problem becomes infeasible at some iteration, then the full list of elements in $\mathrm{MinMG}_{\mathcal{A}}(\mathcal{B})$ has been obtained; otherwise, a new constraint is incorporated, and the problem is solved again.

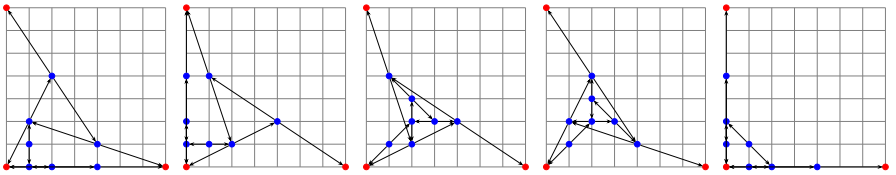


Fig. 2 The entire set $\mathrm{MinMG}_{\mathcal{A}}(\mathcal{B})$ of Example 3.1. Red dots denote the elements of $\mathcal{A}$, whereas blue dots denote the vertices belonging to the mediated sets but not to $\mathcal{A}$

Example 3.1 Let us consider the set $\mathcal{A} = \{(0,0),(7,0),(0,7)\}$ and $\mathcal{B} = \{(1,1)\}$. In Figure 2, we show the five elements in $\mathrm{MinMG}_{\mathcal{A}}(\mathcal{B})$ in case $\mathbb{M} = \mathbb{Z}^2$. They were obtained in the same order as they are plotted by cutting off the solutions previously obtained. All of them have 10 vertices. Observe that although the first and second, and the third and fourth mediated graphs are symmetric with respect to the line $\{x_1 = x_2\}$, the three obtained *proper* mediated graphs have a very different combinatorial structure.

Explicit SOS Decomposition of SONC Polynomials

Let $\mathbb{R}[\boldsymbol{x}] = \mathbb{R}[x_1,\ldots,x_d]$ denote the ring of real d-variate polynomials. For $\boldsymbol{\alpha} = (\alpha_1,\ldots,\alpha_d) \in \mathbb{Z}_+^d$, $\boldsymbol{x}^{\boldsymbol{\alpha}} := x_1^{\alpha_1}\cdots x_d^{\alpha_d}$. Assume that $\mathcal{A} \subset (2\mathbb{Z})^d$ is an affinely independent set. A polynomial $f = \sum_{\boldsymbol{\alpha}\in\mathcal{A}} c_{\boldsymbol{\alpha}}\boldsymbol{x}^{\boldsymbol{\alpha}} + c\boldsymbol{x}^{\boldsymbol{\beta}} \in \mathbb{R}[\boldsymbol{x}]$ is called a *circuit polynomial* (*circuit*, for short) if $c_{\boldsymbol{\alpha}} > 0$, $c \neq 0$, and $\boldsymbol{\beta} \in \mathrm{Conv}(\mathcal{A})^\circ$ [15]. Let $\lambda_{\boldsymbol{\alpha}}$ be the barycentric coordinates of $\boldsymbol{\beta}$ with respect to $\mathcal{A}$. The *circuit number* Θ_f is defined as

$$\Theta_f = \prod_{\boldsymbol{\alpha}\in\mathcal{A}} \left(\frac{c_{\boldsymbol{\alpha}}}{\lambda_{\boldsymbol{\alpha}}}\right)^{\lambda_{\boldsymbol{\alpha}}}.$$

The nonnegativity of the circuit polynomial f can be checked by its circuit number: f is nonnegative if and only if either $|c| \leq \Theta_f$ and $\boldsymbol{\beta} \notin (2\mathbb{Z})^d$, or $c \geq -\Theta_f$ and $\boldsymbol{\beta} \in (2\mathbb{Z})^d$.

Let $\mathcal{B} \subset \mathbb{Z}^d \cap \mathrm{Conv}(\mathcal{A})^\circ$ be a finite set, a polynomial $f \in \mathbb{R}[\boldsymbol{x}]$ is a sum of nonnegative circuits (SONC) polynomial if $f = \sum_{\boldsymbol{\beta}\in\mathcal{B}} f_{\boldsymbol{\beta}}$ where $f_{\boldsymbol{\beta}}$ are nonnegative circuit polynomials supported on $\mathcal{A}\cup\{\boldsymbol{\beta}\}$. Given a mediated graph $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$, the construction of a sum of squares (SOS) decomposition, in fact, sum of binomial squares (SOBS), is based on mediated graphs in the intersection of $\mathcal{M}_{\mathcal{A}}(\boldsymbol{\beta})$ and the set of submediated graphs of G for each $\boldsymbol{\beta} \in \mathcal{B}$. In this setting, each choice of mediated graphs $G_{\boldsymbol{\beta}} = (V_{\boldsymbol{\beta}}, A_{\boldsymbol{\beta}}) \in \mathcal{M}_{\mathcal{A}}(\boldsymbol{\beta})$ that is a subgraph of G for $\boldsymbol{\beta} \in \mathcal{B}$ defines an SOS decomposition of $f = \sum_{\boldsymbol{\beta}\in\mathcal{B}} f_{\boldsymbol{\beta}}$ as $f = \sum_{\boldsymbol{\beta}\in\mathcal{B}} \sigma_{\boldsymbol{\beta}}$ where $f_{\boldsymbol{\beta}} = \sigma_{\boldsymbol{\beta}}$ is an SOS decomposition of $f_{\boldsymbol{\beta}}$ whose number of squares is $|V_{\boldsymbol{\beta}}| - (d+1)$.

However, the authors of this paper note that this claim on the number of squares follows directly from [15, Theorem 5.2] in the extremal case, when the coefficient of

$\boldsymbol{\beta}$ in $f_{\boldsymbol{\beta}}$ is $\pm\Theta_{f_{\boldsymbol{\beta}}}$, since the inner case is derived from a convexity argument applied to the extremal cases. Although the theoretical result characterizing the intersection of SONC and SOS cones by means of the combinatorial structure of the Newton polytope is elegant and complete, the authors of this paper consider it essential to demonstrate an explicit SOS decomposition of SONC polynomials that uses the original support and coefficients of the polynomial.

First, we define the global minimizer of a nonnegative circuit $\boldsymbol{s}_f^* \in \mathbb{R}^d$ as the unique vector satisfying

$$\prod_{k=1}^{d}\left(e^{s_{fk}^*}\right)^{\alpha(j)_k-\alpha(0)_k} = e^{\langle \boldsymbol{s}_f^*, \overrightarrow{\boldsymbol{\alpha}(0)\boldsymbol{\alpha}(j)}\rangle} = \frac{\lambda_j c_0}{\lambda_0 c_j},$$

for all $j = 1, \ldots, d$, where $\mathcal{A} = \{\boldsymbol{\alpha}(0), \ldots, \boldsymbol{\alpha}(d)\}$.

Theorem 3.4 *Let $\mathcal{A} = \{\boldsymbol{\alpha}(0), \ldots, \boldsymbol{\alpha}(d)\}$, and $f(\boldsymbol{x}) = \sum_{j=0}^{d} c_j \boldsymbol{x}^{\boldsymbol{\alpha}(j)} + c\boldsymbol{x}^{\boldsymbol{\beta}}$ be a nonnegative circuit polynomial and let $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\boldsymbol{\beta})$ be a mediated graph. Then,*

1. *if $\boldsymbol{\beta} \notin (2\mathbb{Z})^d$,*

$$f(\boldsymbol{x}) = c_{\boldsymbol{\beta}\boldsymbol{\beta}}\frac{c_0}{\lambda_0}e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\alpha}(0)\rangle}\sum_{m=0}^{1}\frac{\Theta_f + (-1)^m c}{2\Theta_f}\left(\frac{\boldsymbol{x}^{\frac{\boldsymbol{\beta}'}{2}}}{\sqrt{e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\beta}'\rangle}}} + (-1)^m\frac{\boldsymbol{x}^{\frac{\boldsymbol{\beta}''}{2}}}{\sqrt{e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\beta}''\rangle}}}\right)^2$$
$$+ \frac{c_0}{\lambda_0}e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\alpha}(0)\rangle}\sum_{\boldsymbol{\gamma}\in V\setminus(\mathcal{A}\cup\{\boldsymbol{\beta}\})} c_{\boldsymbol{\beta}\boldsymbol{\gamma}}\left(\frac{\boldsymbol{x}^{\frac{\boldsymbol{\gamma}'}{2}}}{\sqrt{e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\gamma}'\rangle}}} - \frac{\boldsymbol{x}^{\frac{\boldsymbol{\gamma}''}{2}}}{\sqrt{e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\gamma}''\rangle}}}\right)^2.$$

2. *if $\boldsymbol{\beta} \in (2\mathbb{Z})^d$ and $c < 0$,*

$$f(\boldsymbol{x}) = (\Theta_f + c)\left(\boldsymbol{x}^{\frac{\boldsymbol{\beta}}{2}}\right)^2 + \frac{c_0}{\lambda_0}e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\alpha}(0)\rangle}\sum_{\boldsymbol{\gamma}\in V\setminus\mathcal{A}} c_{\boldsymbol{\beta}\boldsymbol{\gamma}}\left(\frac{\boldsymbol{x}^{\frac{\boldsymbol{\gamma}'}{2}}}{\sqrt{e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\gamma}'\rangle}}} - \frac{\boldsymbol{x}^{\frac{\boldsymbol{\gamma}''}{2}}}{\sqrt{e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\gamma}''\rangle}}}\right)^2.$$

where $\boldsymbol{\gamma}', \boldsymbol{\gamma}'' \in \delta^+(\boldsymbol{\gamma})$, $c_{\boldsymbol{\beta}\boldsymbol{\gamma}} \geq 0$ is the entry $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ in $(L^+(G)_{V\setminus\mathcal{A}})^{-1}$ for every $\boldsymbol{\gamma} \in V\setminus\mathcal{A}$.

For the sake of the reader, the proof of Theorem 3.4 can be found in the Appendix.

In both cases, the number of squares is $|V| - d$. Then, let $\mathcal{A} \subset (2\mathbb{Z})^d$ be an affine independent set, $\mathcal{B} \subset \mathbb{Z}^d \cap \mathrm{Conv}(\mathcal{A})^\circ$ be a finite set, and let $f \in \mathbb{R}[\boldsymbol{x}]$ be a SONC supported on $\mathcal{A} \cup \mathcal{B}$. Given a mediated graph $G = (V, A) \in \mathcal{M}_{\mathcal{A}}(\mathcal{B})$, and a family of mediated graphs $G_{\boldsymbol{\beta}} = (V_{\boldsymbol{\beta}}, A_{\boldsymbol{\beta}}) \in \mathcal{M}_{\mathcal{A}}(\boldsymbol{\beta})$ that are subgraphs of G for $\boldsymbol{\beta} \in \mathcal{B}$, then $f = \sum_{\boldsymbol{\beta}\in\mathcal{B}} f_{\boldsymbol{\beta}}$ where $f_{\boldsymbol{\beta}}$ is a nonnegative circuit polynomial supported on

$\mathcal{A} \cup \{\boldsymbol{\beta}\}$ can be written as $f = \sum_{\boldsymbol{\beta} \in \mathcal{B}} \sigma_{\boldsymbol{\beta}}$ where $f_{\boldsymbol{\beta}} = \sigma_{\boldsymbol{\beta}}$ is the SOS decomposition of Theorem 3.4. Notice that the involved binomials can be $\left(s_{\boldsymbol{\gamma}} \boldsymbol{x}^{\frac{\boldsymbol{\gamma}'}{2}} - t_{\boldsymbol{\gamma}} \boldsymbol{x}^{\frac{\boldsymbol{\gamma}''}{2}}\right)$ and $\left(s_{\boldsymbol{\beta}} \boldsymbol{x}^{\frac{\boldsymbol{\beta}'}{2}} + t_{\boldsymbol{\beta}} \boldsymbol{x}^{\frac{\boldsymbol{\beta}'}{2}}\right)$ for $\boldsymbol{\gamma} \in V \backslash \mathcal{A}$, $\boldsymbol{\gamma}', \boldsymbol{\gamma}'' \in \delta^+(\boldsymbol{\gamma})$, $s_{\boldsymbol{\gamma}}, t_{\boldsymbol{\gamma}} \in \mathbb{R}_+$ if $\boldsymbol{\beta} \notin (2\mathbb{Z})^d$; and $\left(\boldsymbol{x}^{\frac{\boldsymbol{\beta}}{2}}\right)$ if $\boldsymbol{\beta} \in (2\mathbb{Z})^d$ so the number of squares of the decomposition is less than or equal to $|V| + |\mathcal{B}| - d - 1$, being equality if the coefficient of $\boldsymbol{\beta}$ in $f_{\boldsymbol{\beta}}$ is different from $\pm\Theta_{f_{\boldsymbol{\beta}}}$ for every $\boldsymbol{\beta} \in \mathcal{B}$. The number of squares decreases by one unit for each $\boldsymbol{\beta} \in \mathcal{B}$ that satisfies the equality. Hence, let $(\boldsymbol{x}^*_f, \boldsymbol{y}^*_f)$ be the solution of D-MinMGP for $\mathcal{A}$ and $\mathcal{B}$, defining $V^*_f := \{\boldsymbol{\gamma} \in \mathbb{Z}^d : x^*_{f\boldsymbol{\gamma}} = 1\}$ and $A^*_f := \{(\boldsymbol{\gamma}, \boldsymbol{\gamma}') \in \mathbb{Z}^d \times (2\mathbb{Z})^d : y^*_{f\boldsymbol{\gamma},\boldsymbol{\gamma}'} = 1\}$, then by Theorem 3.3, the mediated graph $G^*_f := (V^*_f, A^*_f) \in \text{MinMG}_{\mathcal{A}}(\mathcal{B})$. Thus, we can apply Theorem 3.4 to each $f_{\boldsymbol{\beta}}$ with G^*_f providing an SOS decomposition of f which reduces the number of squares and the support of the decomposition, increasing the sparsity and reducing the rank of the associated positive semidefinite Gram matrix of f.

3.2 Maximal Mediated Graphs

In what follows, we analyze maximal mediated graphs, i.e., those mediated graphs for $\mathcal{A}$ that have the maximum vertex cardinality. Note that in the *continuous* case, the set of potential vertices for the mediated graph is not finite. Thus, in this section, we focus on maximal mediated graphs for discrete domains. In discrete domains, the number of vertices of any $\mathcal{A}$-mediated graph is at most $|\text{Conv}(\mathcal{A}) \cap \mathbb{M}| < \infty$, which ensures the existence of maximal elements.

Definition 3.2 *(Maximal Mediated Graph)* Let $\mathcal{A} \subset \mathbb{M}$ be a finite set. $G = (V, A) \in \mathcal{M}_{\mathcal{A}}$ is a *maximal $\mathcal{A}$-mediated graph*, if there does not exists $G' = (V', A') \in \mathcal{M}_{\mathcal{A}}$ such that $|V'| > |V|$.

We denote by $\text{MaxMG}_{\mathcal{A}}$ the family of maximal $\mathcal{A}$-mediated graphs. Notice that a maximal mediated set is a maximal element in the poset $(\mathcal{M}_{\mathcal{A}}, \preceq)$.

The first observation that we address is the unification of the notation for maximal mediated sets. The maximal $\mathcal{A}$-mediated set was defined as the $\mathcal{A}$-mediated set that contains every $\mathcal{A}$-mediated set [13], i.e., maximality understood by the order induced by the set inclusion. Indeed, this kind of set is maximum element under this order among all $\mathcal{A}$-mediated sets. In what follows we state the equivalence between our definition of maximal mediated graph and the definition of maximal mediated sets in the literature.

Theorem 3.5 *$G = (V, A) \in \mathcal{M}_{\mathcal{A}}$ is a maximal mediated graph if and only if V is the maximal mediated set. Furthermore, if $G = (V, A)$ and $G' = (V', A')$ are maximal, then $V = V'$.*

Proof Let G be a maximal mediated graph. If there is an $\mathcal{A}$-mediated set V' such that $V' \nsubseteq V$, then $V \subset V \cup V'$. Since $V \cup V'$ is an $\mathcal{A}$-mediated set, there exists an

$\mathcal{A}$-mediated graph $G' = (V \cup V', A')$. However, $G' \succ G$ and that is not possible by the maximality of G. So, V is the maximal mediated set.

On the other hand, let V be the maximal $\mathcal{A}$-mediated set. Assume there exists $G' = (V', A') \succeq G$, this implies either $|V'| > |V|$ or $G' = G$. However, the first condition cannot be true for the maximality of V, so $G' = G$. Hence, G is maximal. □

In what follows, we provide our mathematical optimization formulation that we propose to compute the maximal mediated graph for a set $\mathcal{A}$.

Theorem 3.6 *Let $\mathcal{A} \subset \mathbb{Z}^d$ be a finite set of points. The following integer linear programming model allows us to compute a mediated graph in* $\mathrm{MaxMG}_{\mathcal{A}}$.

$$
\begin{aligned}
\text{Maximize} \quad & |\mathcal{A}| + \sum_{\boldsymbol{v} \in \mathcal{V} \setminus \mathcal{A}} x_{\boldsymbol{v}} && && \text{(D-MaxMGP)}\\
\text{subject to} \quad & \sum_{\boldsymbol{v} \in \mathcal{V}} y_{\boldsymbol{v}\boldsymbol{w}} \leq |\mathcal{V}| x_{\boldsymbol{v}}, && && \forall \boldsymbol{w} \in \mathcal{V},\\
& \sum_{\boldsymbol{w} \in \mathcal{V}} y_{\boldsymbol{v}\boldsymbol{w}} \leq |\mathcal{V}| x_{\boldsymbol{v}}, && && \forall \boldsymbol{v} \in \mathcal{V},\\
& y_{\boldsymbol{v}(2\boldsymbol{v}-\boldsymbol{w})} \geq y_{\boldsymbol{v}\boldsymbol{w}}, && && \forall \boldsymbol{v} \neq \boldsymbol{w} \in \mathcal{V} \text{ if } 2\boldsymbol{v} - \boldsymbol{w} \in \mathcal{V},\\
& y_{\boldsymbol{v}\boldsymbol{w}} = 0, && && \forall \boldsymbol{v} \neq \boldsymbol{w} \in \mathcal{V} \text{ if } 2\boldsymbol{v} - \boldsymbol{w} \notin \mathcal{V},\\
& \sum_{\boldsymbol{w} \in \mathcal{V}} y_{\boldsymbol{v}\boldsymbol{w}} = 2x_{\boldsymbol{v}}, && && \forall \boldsymbol{v} \in \mathcal{V} \setminus \mathcal{A},\\
& x_{\boldsymbol{v}} = 1, && && \forall \boldsymbol{v} \in \mathcal{A},\\
& x_{\boldsymbol{v}} \in \{0, 1\}, && && \forall \boldsymbol{v} \in \mathcal{V},\\
& y_{\boldsymbol{v}\boldsymbol{w}} \in \{0, 1\}, && && \forall \boldsymbol{v}, \boldsymbol{w} \in \mathcal{V},
\end{aligned}
$$

where $\mathcal{V} = \mathrm{Conv}(\mathcal{A}) \cap \mathbb{Z}^d$.

Proof Note that the formulation is similar to the one provided in Theorem 3.3. Instead of minimizing the number of vertices in the obtained mediated graph, this cardinality is maximized. □

Intersection of SONC and SOS Cones

The main application of maximal mediated graphs is the certification that a SONC (sums of nonnegative circuits) polynomial is sum-of-squares (SOS) decomposable. In case the polynomial is SOS, one can incorporate the constraints induced by such a polynomial as SDP constraints in a nonlinear optimization problem, and then efficiently handle them using available solvers. Nevertheless, no polynomial-time algorithm is currently known for certifying the SOS condition in general [21]. Our approach allows for an efficient and automated methodology to check, in practice, this condition for these structured polynomials. In what follows, we provide further details about the application of our methodology to this problem.

Table 1 Sample sizes of the simplices from [13]

d	2	2	2	3	3	4	4	4	4	4	5	5	6	6	7	7	8	9
m	50	100	150	10	16	6	8	10	14	16	8	16	16	20	4	16	16	16
#	1000	1000	300	500	1000	150	1000	1000	1000	1000	1000	1000	1000	1000	15	1000	1000	300

Let $\mathcal{A} \subset (2\mathbb{Z})^d$ be a finite affine independent set, and let $\mathcal{B} \subset \mathbb{Z}^d \cap \text{Conv}(\mathcal{A})^\circ$ be a finite set. The authors in [13] showed a complete characterization of the polynomials supported on $\mathcal{A} \cup \mathcal{B}$ that lie in the intersection of SONC and SOS cones. That result can be adapted in terms of the solution of D-MaxMGP.

Corollary 3.3 *Let $f \in \mathbb{R}[\boldsymbol{x}]$ be a* SONC *supported on $\mathcal{A} \cup \mathcal{B}$ and $\boldsymbol{x}^*$ the solution of* (D-MaxMGP) *for $\mathcal{A}$. Then, f is* SOS *if and only if for every $\boldsymbol{\beta} \in \mathcal{B}$ either $\boldsymbol{x}^*_{\boldsymbol{\beta}} = 1$ or $\boldsymbol{\beta} \in (2\mathbb{Z})^d$ and its coefficient is positive.*

Proof The result is straightforward from [13, Theorem 3.9] and Theorem 3.6. □

4 Computational Experiments

In this section, we report the results of our computational experiments, which we performed to validate the proposed approaches and compare them with other approaches in the literature.

4.1 Data

We use the datasets provided in [13] and that the authors made publicly available at https://polymake.org/downloads/MMS/csv/. The dataset contains simplicial sets, $\mathcal{A} = \{\boldsymbol{v}_0, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_d\}$, in dimensions $d \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ of different sizes, measured by their degrees:

$$m(\mathcal{A}) := \min \left\{ \rho \in \mathbb{Z}_+ : \sum_{l=1}^{d} w_l \leq 2\rho \text{ for all } \boldsymbol{w} \in \text{Conv}(\mathcal{A}) \cap (2\mathbb{Z})^d \right\}.$$

The authors generated all the simplicial sets and lattices for some of the dimensions and degrees and sampled these sets for the rest of the dimensions and higher degrees. In total, 8, 941, 852 maximal mediated sets were computed in [13], distributed by dimension as shown in Table 1. We randomly selected some of these instances to compute the minimal and maximal mediated graph using our approach. In the third column of Table 1, we detail the number of random instances selected for each combination of d and m.

We compute the maximal mediated graphs of the selected random sets $\mathcal{A}$, as detailed above. We also run the enumeration-and-filtering algorithm proposed in [13] to compare the efficiency of our approach with respect to the only previous proposal for this task.

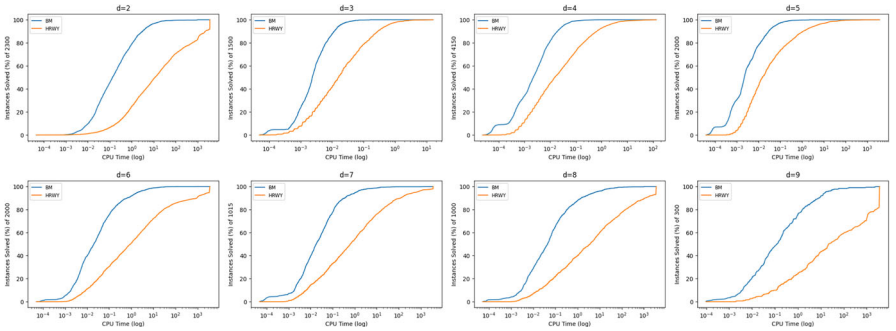


Fig. 3 Performance profiles for our approach (blue) and the one in [13] (orange)

For the minimal mediated graphs, a set of interior points $\mathcal{B}$ is required to obtain *proper* subgraphs. We then run our model for a random set of $s \in \{1, 3, 5\}$ integer points inside the convex hull of the points in $\mathcal{A}$ to analyze the performance on different numbers of points in $\mathcal{B}$.

The models were coded in `Python` and solved using the optimization solver `Gurobi 12.0` on an Apple M1 Max with 64 GB RAM.

4.2 Results on Maximal Mediated Graphs

We ran our MILP model for all the samples of the instances provided in Table 1. In total, 15, 110 problems were solved. We also run the algorithm proposed in [13, Algorithm 4.3].

In Figure 3, we show the performance profiles for obtaining the maximal mediated graph of both our mathematical optimization-based approach and the algorithm proposed in [13] (named `BM` and `HRWY`, for the initials of the authors' last names). In the plots, we represent the percent of instances solved within a given unit of CPU time (using log-scales to ease the distinction). For a fair comparison, we do not include in those times the time required to enumerate the points inside the given lattices. It is evident that the performance of our approach is superior to that of [13]. Although solving a MILP may require, in the worst case, the enumeration of the whole set of integer feasible points, the solution methods for these problems are designed to avoid such an enumeration by branching, bounding, and reducing the feasible region by cutting off part of the space.

The summary of the obtained results is shown in Table 2, where we report, for each value of d and m, the average CPU times (in seconds) with both our methodology (`time_BM`) and the enumerative methodology (`time_HRWY`) only for those instances that were optimally solved separately with each procedure. The percent of instances in each row that were not optimally solved within one-hour time limit with the enumerative methodology is reported in column `unsolved_HRWY`. Our methodology was capable of solving all the instances optimally within the time limit.

In Figure 4, we show for each dimension d the average CPU times (log-scale) to compute the maximal mediated graph with the optimization-based methodology

Table 2 Summary of our computational experiments for the maximal mediated graph

d	m	time_BM	time_HRWY	unsolved_HRWY
2	**50**	0.05	5.71	0%
	100	1.97	373.61	4.30%
	150	23.91	818.49	46.67%
3	**10**	0.00	0.01	0%
	16	0.01	0.20	0%
4	**6**	0.00	0.00	0%
	8	0.00	0.00	0%
	10	0.00	0.04	0%
	14	0.01	0.83	0%
	16	0.02	1.63	0%
5	**8**	0.00	0.02	0%
	16	0.04	4.60	0.10%
6	**16**	0.09	32.05	0.30%
	20	2.09	197.51	9.90%
7	**4**	0.00	0.01	0%
	16	3.16	58.28	2.20%
8	**16**	5.04	130.39	6.80%
9	**16**	22.51	307.61	17.67%

that we propose (BM) and with the enumerative strategy proposed in [13] (HRWY). We also represent in the same plot the average CPU times required to enumerate the points inside the simplex, required to compute the optimal mediated graph with both methodologies.

In view of the results, we conclude that the CPU time required by our optimization-based methodology to obtain the maximal mediated graphs is significantly smaller than that required by the enumerative procedure proposed in [13] and, in some cases, it is even able to construct the optimal solutions when the enumerative approach is not capable of doing so.

Although the construction involved in the enumerative approach can be useful for understanding the geometry of the mediated graphs, our approach has the advantage that it allows us to solve larger instances in reasonable CPU time. The combination of our approach and the reduction strategies in the enumerative approach by incorporating conditions in the form of linear inequalities in our model could result in a better approach, which will be explored in a forthcoming paper.

4.3 Results for Minimal Mediated Graphs

For the minimal mediated graphs, we randomly select 20 instances from each row, i.e., for each dimension d. For each of them, we randomly select s interior points of the simplex $\mathcal{A}$, for $s \in \{1, 3, 5\}$. These points form the set $\mathcal{B}$. Then, we consider three different domains for the vertices in the minimal mediated graph $\mathbb{M} \in \{\mathbb{R}^d, \mathbb{Z}^d, (2\mathbb{Z})^d\}$.

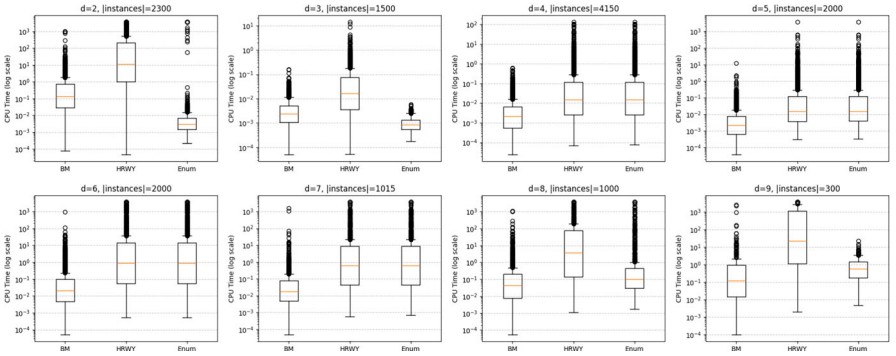


Fig. 4 Boxplots for the CPU times (on a log-scale) required to compute the maximal mediated graphs with our methodology (BM), the methodology in [13], and the CPU time required to enumerate the required points

In total, we solve 270 instances with each of the three optimization models that we propose. We run the extended approach highlighted in Remark 3.2 to construct the whole set of minimal mediated graphs for given sets $\mathcal{A}$ and $\mathcal{B}$ and a specific domain $\mathbb{M}$.

For $\mathbb{M} = \mathbb{R}^d$ (continuous domain), since the mathematical optimization problem is affected by the number of potential mediated vertices in the graph (upper bounded by $\nu_{\mathcal{A}}(\mathcal{B})$, which can be large), following the suggestion in [5], we solve the problems iteratively. Specifically, we start by taking $\mathcal{V}$ to be an index set with cardinality 1, and solve the problem. In case the problem is feasible, we are done, and the solution is a minimal mediated graph. Otherwise, we increase the cardinality of $\mathcal{V}$ and repeat the process until a feasible solution is found. In most cases, the cardinal of the minimal mediated graph is much smaller than the upper bound, and we avoid solving huge integer linear problems. In general, we empirically tested that certifying infeasibility requires much less time than solving a single but larger integer linear problem.

In Table 3, we summarize the results of our experiments. There, the first two columns indicate the dimension (d) and the number of random points chosen in the set $\mathcal{B}$. Then, for each of the three domains $\mathbb{M} \in \{\mathbb{R}^d, \mathbb{Z}^d, (2\mathbb{Z})^d\}$, we report the average sizes of the minimal mediated graph, the CPU time (in seconds) required to compute one minimal mediated graph, the average CPU time (in seconds) per extra minimal mediated graph computed, and the number of minimal mediated graphs obtained. For the domain $(2\mathbb{Z})^d$, either a single minimal mediated set was found or the problem was infeasible. Thus, we do not report the value *Av. Time Rest* for this domain since all the values are zero.

As expected, as the domain is larger, the problem becomes more challenging. Although one could also expect that the larger the size of $|\mathcal{B}|$, the more difficult the problem, this is not always true, as can be seen for $d = 2$, where the problem for $s = 3$ is more time demanding than the others. The same happens for the computation of more than one minimal mediated graph in $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$.

Regarding the size of the obtained minimal mediated graph, the number of vertices is slightly larger for $(2\mathbb{Z})^d$ than for $\mathbb{Z}^d$, and the same happens for $\mathbb{Z}^d$ and $\mathbb{R}^d$. Note that

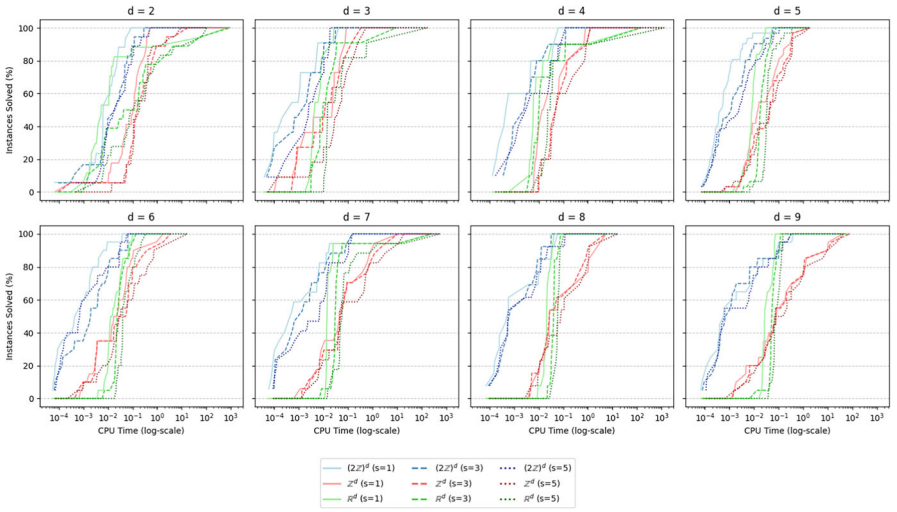


Fig. 5 Performance profile for our experiments on the minimal mediated graphs. Blue lines represent $\mathbb{M} = (2\mathbb{Z})^d$, green lines $\mathbb{M} = \mathbb{R}^d$ and red lines $\mathbb{M} = \mathbb{Z}^d$. Solid lines indicate the case $s = 1$, dashed lines $s = 3$, and dotted lines $s = 5$

although the approaches are different, they all solve the same problems in different, nested feasible regions. In fact, feasible solutions in the domain $(2\mathbb{Z})^d$ are also feasible for $\mathbb{Z}^d$, and those for $\mathbb{Z}^d$ are feasible for $\mathbb{R}^d$. This observation also affects the number of graphs in $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$, where for the domain $\mathbb{R}^d$ a larger numbers of minimal mediated graphs was found.

Note that the case $d = 2$ is particular, since in the dataset in [13], the authors consider simplices with extreme points with large coordinates. Thus, the planar case can be more time-consuming than the higher-dimensional case, because of the degrees of the tested simplices.

In Fig. 5, we show the performance profiles (on a log-scale to ease their reading) for our experiments on the minimal mediated graph distinguished by dimension d, size of the set $\mathcal{B}$, s, and domain. Each line was constructed by indicating the percent of instances optimally solved in less than a given number of seconds (on a log-scale). The least time-consuming approach (upper lines) seems to be the domain $(2\mathbb{Z})^d$ for $s = 1$ (solid blue lines). In contrast, the most challenging problems were the continuous domain (green) and some of those in the $\mathbb{Z}^d$ domain (red).

In conclusion, our approach to computing minimal mediated graphs was capable of solving all the instances (even the $d = 9$ ones) in more than reasonable CPU time, b both for computing a single mediated graph and for computing all the graphs in the family $\text{MinMG}_{\mathcal{A}}(\mathcal{B})$, for all the domains that we consider. Thus, this novel approach with multiple applications in conic decomposition has been proven to be useful and can be practically applied to these problems.

Table 3 Summary of the results of our experiments for the minimal mediated graph

d	$\lvert\mathcal{B}\rvert$	# Vertices			Time First Sol.			Avg. Time Rest		$\lvert\text{MinMG}_{\mathcal{A}}(\mathcal{B})\rvert$		
		$(2\mathbb{Z})^d$	$\mathbb{Z}^d$	$\mathbb{R}^d$	$(2\mathbb{Z})^d$	$\mathbb{Z}^d$	$\mathbb{R}^d$	$\mathbb{Z}^d$	$\mathbb{R}^d$	$(2\mathbb{Z})^d$	$\mathbb{Z}^d$	$\mathbb{R}^d$
2	1	5.24	5.24	4.53	0.02	0.15	43.32	0.32	0.01	1.00	1.06	1.29
	3	7.83	7.78	7.06	0.04	1.35	55.53	4.86	2.51	1.00	2.94	17.56
	5	10.39	10.11	8.94	0.07	0.91	9.78	1.91	0.30	1.00	6.44	17.28
3	1	5.18	5.18	5.18	0.01	0.03	0.01	0.02	0.00	1.00	1.00	1.00
	3	8.00	7.73	7.73	0.01	0.06	0.86	0.11	0.29	1.00	1.45	2.64
	5	10.55	10.45	10.45	0.01	0.11	16.10	0.07	3.62	1.00	3.36	10.73
4	1	6.00	6.00	5.40	0.01	0.18	13.83	0.26	0.00	1.00	1.00	1.00
	3	8.60	8.60	7.80	0.02	0.26	15.32	0.33	0.02	1.00	1.30	1.40
	5	10.40	10.40	9.40	0.03	0.33	131.70	0.33	0.01	1.00	1.40	1.40
5	1	7.13	7.13	7.13	0.00	0.12	0.01	0.18	0.00	1.00	1.00	1.00
	3	9.39	9.26	9.26	0.01	0.15	0.02	0.20	0.01	1.00	1.10	1.19
	5	11.10	11.39	11.39	0.01	0.18	0.05	0.24	0.04	1.00	1.16	1.32
6	1	8.10	8.10	8.10	0.00	0.16	0.03	0.31	0.00	1.00	1.00	1.00
	3	10.25	10.25	10.25	0.01	0.32	0.04	0.35	0.01	1.00	1.00	1.00
	5	11.85	12.50	12.50	0.01	1.15	0.08	0.64	0.06	1.00	1.20	1.40
7	1	9.00	9.00	8.47	0.01	0.87	14.69	1.68	0.00	1.00	1.00	1.00
	3	11.06	11.06	10.41	0.02	1.01	15.66	1.63	0.01	1.00	1.00	1.00
	5	13.47	13.47	12.71	0.03	2.11	33.70	2.55	0.13	1.00	1.18	1.82
8	1	10.08	10.08	10.08	0.01	0.81	0.02	1.19	0.00	1.00	1.00	1.00
	3	12.08	12.08	12.08	0.01	0.63	0.04	1.13	0.01	1.00	1.08	1.08
	5	14.23	14.23	14.23	0.01	1.55	0.06	3.53	0.02	1.00	1.08	1.08
9	1	11.00	11.00	11.00	0.03	3.36	0.04	10.86	0.00	1.00	1.00	1.00
	3	13.05	13.05	13.05	0.03	5.30	0.06	11.15	0.01	1.00	1.00	1.00
	5	15.10	15.10	15.10	0.03	4.32	0.09	8.43	0.02	1.00	1.00	1.00

5 Conclusions

We introduced the notion of *mediated graphs* as a unifying combinatorial framework motivated by recent developments in convex and conic geometry. By bridging discrete graph-based structures and continuous conic optimization, we developed a structural theory for mediated graphs, analyzed their geometric properties, and defined a natural partial order that allows the characterization of extremal mediated graphs, namely minimal and maximal mediated graphs.

A key contribution of this work is the formulation of mixed-integer linear optimization models for computing minimal and maximal mediated graphs on arbitrary domains. These models enable exact and systematic constructions, overcoming the limitations of previous ad hoc approaches. As applications, we derived SOCP representations for generalized power cone programs, explicit SONC certificates for circuit polynomials, and hybrid SOS-SONC constructions, highlighting the relevance of mediated graphs in polynomial and conic optimization.

Finally, computational experiments demonstrate that the proposed optimization-based approach is practical and scalable, allowing the efficient computation of mediated graphs of nontrivial size and dimension.

Future Research. The structural behavior of optimal mediated graphs remains a fertile area for further investigation. The existence of particular families of sets $\mathcal{A}$ and $\mathcal{B}$ for which minimal mediated graphs simplify, or admit closed-form descriptions, is presently unknown. In the case of maximal mediated graphs, Reznick [27] conjectured, and the authors in [13] empirically tested for millions of simplices that in dimension 2 every simplex is either an H-simplex or an M-simplex (meaning that its maximal mediated graph is either the full set of integer points in $\mathrm{Conv}(\mathcal{A})$ or the set $\mathcal{A}$ together with its midpoints). A polyhedral study of the integer linear programs we introduce here may provide new insight into this conjecture.

The impact of these constructions on polynomial optimization, particularly regarding the size of the resulting convex relaxations, remains to be explored. A deeper analysis of these relaxations, in terms of their size, computational complexity, and tightness, constitutes an important direction for future research. The framework developed in this paper provides exactly the structural and algorithmic tools required to pursue this research line, and we plan to examine these implications in subsequent work. This perspective also aligns with current research efforts aimed at finding and constructing sparse bases for representing positive polynomials at a controlled computational cost. Progress in this direction would have a direct impact on the scalability of Moment-SOS hierarchy approaches for tackling challenging global optimization problems.

Finally, motivated by applications in SOS and SOCP representations, exact mediated structures may be computationally inefficient, as they can require overly large graphs. In such settings, it may be advantageous to relax the mediated condition in order to obtain approximate or *near-mediated* decompositions. The role of integer linear programming in designing and analyzing such approximation schemes remains an open question and an intriguing direction for future research.

Proof of Theorem 3.4

Proof of Theorem 3.4 For the first part, notice that the polynomial f can be written as a convex combination of the extremal nonnegative circuit with its support

$$f(\boldsymbol{x}) = \sum_{m=0}^{1} \frac{\Theta_f + (-1)^m c}{2\Theta_f} \left[\sum_{j=0}^{d} c_j \boldsymbol{x}^{\boldsymbol{\alpha}(j)} + (-1)^m \Theta_f \boldsymbol{x}^{\boldsymbol{\beta}} \right]$$

Now, note $\boldsymbol{s}_f^*$ is the solution of

$$\begin{bmatrix} \boldsymbol{\alpha}(1) - \boldsymbol{\alpha}(0) \\ \vdots \\ \boldsymbol{\alpha}(d) - \boldsymbol{\alpha}(0) \end{bmatrix} \boldsymbol{s}_f^* = \begin{bmatrix} \log\left(\frac{\lambda_1 c_0}{\lambda_0 c_1}\right) \\ \vdots \\ \log\left(\frac{\lambda_d c_0}{\lambda_0 c_d}\right) \end{bmatrix}$$

the linear system has a unique solution by affine independence of $\boldsymbol{\alpha}(0), \boldsymbol{\alpha}(1), \ldots, \boldsymbol{\alpha}(d)$. In this way, we have the extremal case $f_m^*(\boldsymbol{x}) = \sum_{j=0}^{d} c_j \boldsymbol{x}^{\boldsymbol{\alpha}(j)} + (-1)^m \Theta_f \boldsymbol{x}^{\boldsymbol{\beta}}, m = 0, 1,$ evaluated in $\left(e^{\boldsymbol{s}_f^*} \circ \boldsymbol{x}\right)$, where $\circ$ stands for the Hadamard product becomes

$$f_m^*\left(e^{\boldsymbol{s}_f^*} \circ \boldsymbol{x}\right) = \frac{c_0}{\lambda_0} e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\alpha}(0) \rangle} \left[\sum_{j=0}^{d} \lambda_j \boldsymbol{x}^{\boldsymbol{\alpha}(j)} + (-1)^m \boldsymbol{x}^{\boldsymbol{\beta}} \right].$$

Now, $f_1^*\left(e^{\boldsymbol{s}_f^*} \circ \boldsymbol{x}\right)$ is a positive multiple of a simplicial agiform, so by Reznick's SOS decomposition of agiforms [27] and the fact that $\boldsymbol{\beta} \notin (2\mathbb{Z})$, it is easy to see that

$$f_m^*\left(e^{\boldsymbol{s}_f^*} \circ \boldsymbol{x}\right) = \frac{c_0}{\lambda_0} e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\alpha}(0) \rangle} \left[c_{\boldsymbol{\beta}\boldsymbol{\beta}} \left(\boldsymbol{x}^{\frac{\boldsymbol{\beta}'}{2}} + (-1)^m \boldsymbol{x}^{\frac{\boldsymbol{\beta}''}{2}} \right)^2 + \sum_{\boldsymbol{\gamma} \in V \setminus (\mathcal{A} \cup \{\boldsymbol{\beta}\})} c_{\boldsymbol{\beta}\boldsymbol{\gamma}} \left(\boldsymbol{x}^{\frac{\boldsymbol{\gamma}'}{2}} - \boldsymbol{x}^{\frac{\boldsymbol{\gamma}''}{2}} \right)^2 \right]$$

where $c_{\boldsymbol{\beta}\boldsymbol{\gamma}} \geq$ is the entry $(\boldsymbol{\beta}, \boldsymbol{\gamma})$ of $(L^+(G)_{V \setminus \mathcal{A}})^{-1}$ for every $\boldsymbol{\gamma} \in V \setminus \mathcal{A}$ (Proposition 2.1.7). So, gathering the terms

$$f(e^{\boldsymbol{s}_f^*} \circ \boldsymbol{x}) = \frac{c_0}{\lambda_0} e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\alpha}(0) \rangle} \sum_{m=0}^{1} \frac{\Theta_f + (-1)^m c}{2\Theta_f} \left[c_{\boldsymbol{\beta}\boldsymbol{\beta}} \left(\boldsymbol{x}^{\frac{\boldsymbol{\beta}'}{2}} + (-1)^m \boldsymbol{x}^{\frac{\boldsymbol{\beta}''}{2}} \right)^2 \right] + \frac{c_0}{\lambda_0} e^{\langle \boldsymbol{s}_f^*, \boldsymbol{\alpha}(0) \rangle} \sum_{\boldsymbol{\gamma} \in V \setminus (\mathcal{A} \cup \{\boldsymbol{\beta}\})} c_{\boldsymbol{\beta}\boldsymbol{\gamma}} \left(\boldsymbol{x}^{\frac{\boldsymbol{\gamma}'}{2}} - \boldsymbol{x}^{\frac{\boldsymbol{\gamma}''}{2}} \right)^2,$$

it just remains to undo the transformation to achieve the desired result.

To prove the second part, notice that the polynomial f can be written as

$$f(\boldsymbol{x}) = \sum_{j=0}^{d} c_j \boldsymbol{x}^{\boldsymbol{\alpha}(j)} - \Theta_f \boldsymbol{x}^{\boldsymbol{\beta}} + (\Theta_f + c)\boldsymbol{x}^{\boldsymbol{\beta}} = f_1^*(\boldsymbol{x}) + (\Theta_f + c)\left(\boldsymbol{x}^{\frac{\boldsymbol{\beta}}{2}}\right)^2.$$

Now, use the same argument for $f_1^*(\boldsymbol{x})$ in the proof of the first part and yield the result.

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Declarations

Competing Interests The authors declare that they have no competing interests.

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